

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL 3)

Assessment Tool Weightage: 10%

Duration : 45 Minutes

QUESTIONS: 2 Total Marks:20

Annexure

A

Descriptive Written Test

Code of Conduct:

I G.Bhavya [192411114] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Problem Understanding

A smart home automation system controls **lighting, temperature, and security** based on user behavior and environmental conditions. It receives real-time inputs such as motion detection, temperature changes, and time of day while learning user preferences to improve comfort, energy efficiency, and security.

Types of Intelligent Agents

1. Simple Reflex Agent

A **Simple Reflex Agent** makes decisions based only on the current input using predefined rules.

Example:

- If motion is detected, turn on the lights.
- If no motion is detected, turn off the lights.

Advantage: Fast and simple.

Limitation: Cannot remember past events or learn user preferences.

2. Model-Based Agent

A **Model-Based Agent** maintains an internal model of the environment and uses previous information to make decisions.

Example:

- If the room was previously occupied, keep the lights on for a few minutes even if no motion is detected.
- Adjust the temperature based on both the current temperature and previous settings.

Advantage: Works well in partially observable environments.

Limitation: More complex than a Simple Reflex Agent.

3. Goal-Based Agent

A **Goal-Based Agent** selects actions to achieve a specific goal.

Example:

- Maintain the room temperature at 24°C.
- Keep the home secure when the user is away.
- Reduce unnecessary energy consumption.

Advantage: Chooses the best action to achieve the desired objective.

Limitation: Requires planning before making decisions.

4. Utility-Based Agent

A **Utility-Based Agent** selects the action that provides the maximum overall benefit.

Example:

- Balance comfort and electricity usage.
- Choose energy-saving mode when electricity consumption is high.
- Maintain security while minimizing unnecessary alarms.

Advantage: Produces the best overall performance.

Limitation: Utility calculation can be complex.

Best Approach

A **Hybrid Agent (Model-Based + Goal-Based + Utility-Based)** is the most effective solution.

Reason:

- The Model-Based Agent remembers user preferences.
- The Goal-Based Agent achieves comfort and security.
- The Utility-Based Agent optimizes energy efficiency and overall performance.

Justification

A hybrid approach provides better comfort, improves security, reduces energy consumption, and adapts to changing user behavior, making it the most suitable intelligent agent for a smart home automation system.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Problem Understanding

A robot is placed in an unknown maze and must find the exit without prior knowledge of the layout. The robot must search the environment while avoiding dead ends and finding the shortest or least-cost path.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the **lowest cumulative cost** first.

Working:

1. Start from the initial position.
2. Expand the lowest-cost node.
3. Update the priority queue with neighbouring nodes.
4. Repeat until the exit is found.

Advantages

- Finds the optimal path.
- Works well when path costs are different.
- Guarantees completeness.

Disadvantages

- Requires more memory.
- Can be slower for large search spaces.

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs Depth-First Search with increasing depth limits.

Working:

1. Search with depth limit 0.
2. Increase the limit to 1, 2, 3, and so on.
3. Continue until the exit is found.

Advantages

- Uses less memory.
- Complete for finite search spaces.
- Suitable when the solution depth is unknown.

Disadvantages

- Repeats node expansions.
- May take more time than BFS.

Time and Space Complexity

Algorithm	Time Complexity	Space Complexity
UCS	$O(b^{(1+C^*/\epsilon)})$	$O(b^{(1+C^*/\epsilon)})$
IDS	$O(b^d)$	$O(bd)$

Where:

- **b** = Branching factor
- **d** = Depth of the solution
- **C*** = Optimal path cost
- **ε** = Minimum step cost

Intelligent Agent

A **Goal-Based Agent** is suitable because its objective is to reach the exit.

A **Model-Based Agent** can remember explored paths and avoid revisiting the same locations.

Effect of Agent Design

The agent design determines how the robot:

- Collects information.
- Chooses the next move.
- Avoids repeated exploration.
- Reaches the goal efficiently.

A better agent design improves decision-making and reduces search time.

Best Combination

Model-Based Goal-Based Agent + Uniform Cost Search (UCS)

Reason:

- The Model-Based Agent stores information about explored paths.
- The Goal-Based Agent focuses on reaching the exit.
- Uniform Cost Search guarantees the least-cost path, making it suitable for an unknown maze where different paths may have different costs.

Justification

The combination of a **Model-Based Goal-Based Agent** and **Uniform Cost Search** provides efficient decision-making, remembers explored areas, adapts to the unknown environment, and finds the optimal path to the exit. It is the most suitable approach for solving the maze problem.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand
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